

E-MultiTool Documentation

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1. Introduction

The E-MultiTool is a multifunctional electronic device designed around an STM32 microcontroller. The project combines embedded software and hardware design to create a tool capable of performing 5 functions: angle measurement, flat level detection, continuity testing, PWM generation and free fall detection. This documentation presents the design and development of the E-MultiTool, including its hardware architecture and the implementation and results of each feature.

2. Hardware/PCB Design

The E-MultiTool was implemented on a two-layer PCB. The bottom layer was primarily used as a ground plane, while the top layer was used to route signal traces. Due to space constraints on the top layer, a small number of signals were routed through sections of the ground plane. It should also be noted that the board does not include dedicated ESD protection circuitry. Therefore, it is recommended to operate this device in an ESD safe area.

2.1 Hardware Architecture

The overall hardware architecture of the E-MultiTool is shown in Figure 1, with the STM32F0 microcontroller serving as the central component. The rotary encoder is connected to TIM3 in encoder mode to detect user input, while its switch is connected to a GPIO interrupt for selecting and exiting the 4 options. Moreover, the continuity test pin (TST) is connected to GPIO pin PA11 and the line is pulled high through a 10k Ω resistor. The onboard accelerometer and OLED display communicate with the MCU through I²C1 and I²C2, respectively. TIM14 is used to generate the PWM output, and an external 16 MHz crystal provides the MCU clock source.

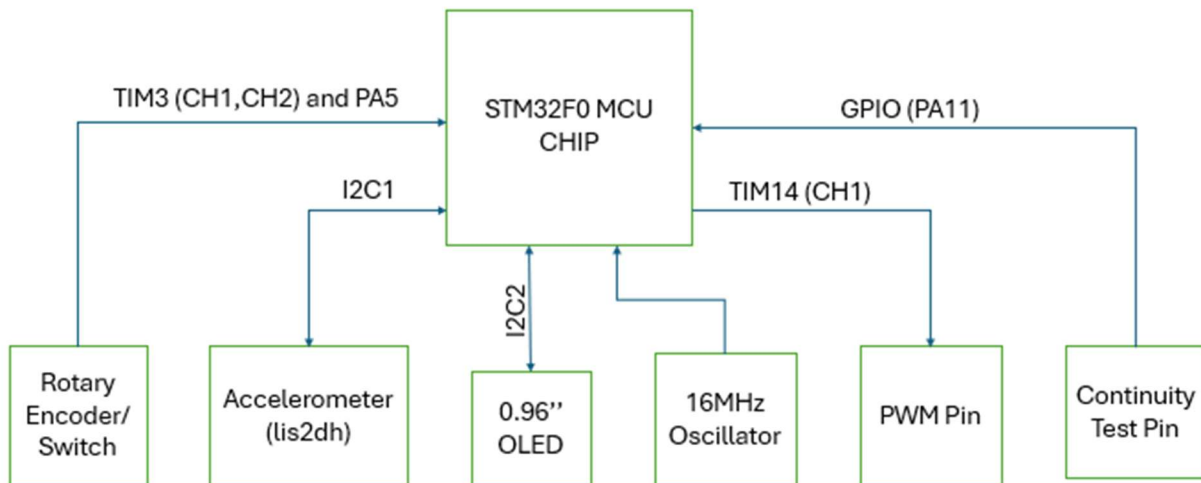


Figure 1: Hardware structure

2.2 STM32F0 Microcontroller

The STM32F0 serves as the central controller of the E-MultiTool, handling sensor interfacing, data processing, and user input. The MCU operates from a 3.3 V supply with a π -filter on the analog supply and 100nF decoupling capacitors on the digital supply pins, supplemented by a 10 μ F bulk capacitor. An external 16 MHz crystal provides the system clock. PA3 drives an onboard LED, while pins PB12–PB15 are reserved for future expansion. The NRST pin is connected to a push button for manual reset, and the board is programmed and debugged through an ST-LINK interface. Figure 2 shows the primary pin connections for the STM32F0 MCU.

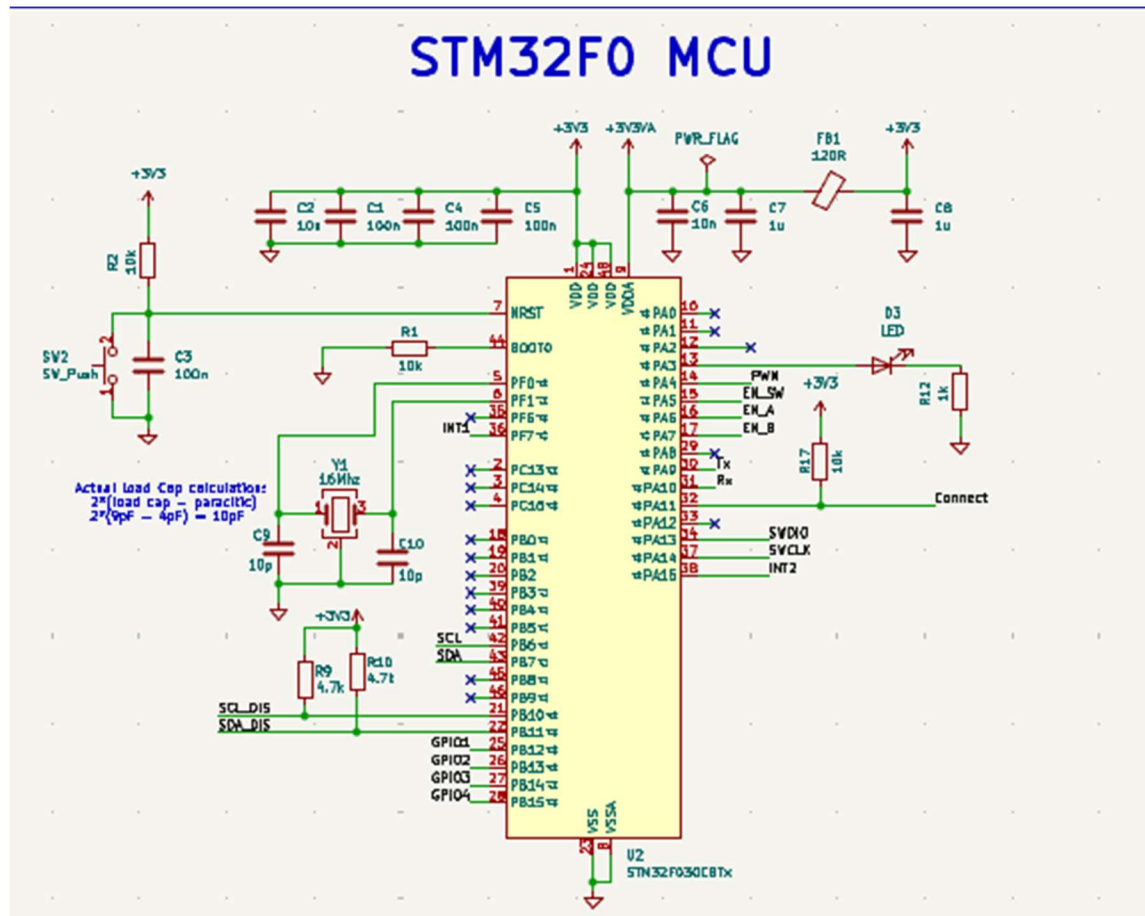


Figure 2: STM32F0 Connections

2.3 Power Supply

The power supply to the E-MultiTool comes from an external power source that is fed through a micro-USB B port. Since the external source produces 5V, a regulator is used to step down the voltage to 3.3V. An LED is used to check if power is being delivered to the board. Figure 3 illustrates the power connection schematic.

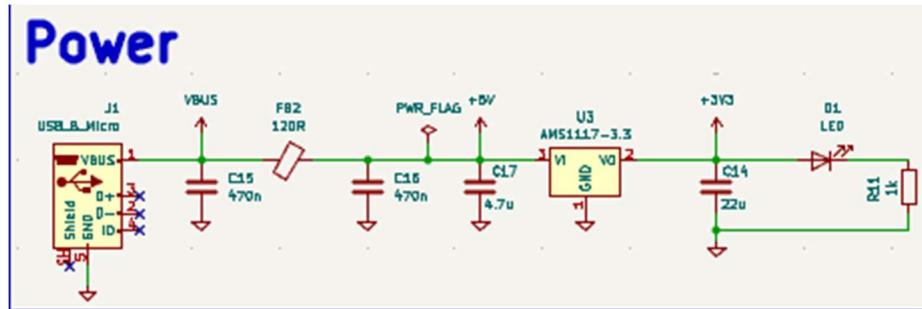


Figure 3: Power supply hardware configuration

2.4 I2C and USART

To communicate with the accelerometer and the OLED, I2C1 and I2C2 are used respectively. The SCL and SDA lines of both I2C is pulled up through a 4.7kΩ resistor. The board also support USART communication to transfer data between microcontrollers if needed.

2.5 Crystal Oscillator and Timers

A 16MHz external crystal is placed close to the MCU to avoid any clock skew. The load capacitance stated in the oscillator's specification was 9pF. But parasitic capacitance was also considered, and the load capacitance was calculated as follows:

$$\text{Load Capacitance} = 2 * (\text{Actual Capacitance} - \text{Parasitic Capacitance}) = 2 * (9\text{pF} - 4\text{pF}) = 10\text{pF}$$

It is to be noted that the 4pF parasitic capacitance is an approximate estimate.

TIM 3 is used in encoder mode where two channels automatically track the rotary encoder's position. By sampling both edges of both channels, one physical turn or click attributes to four counts. Since there are 4 options, a max counter value of 15 is set with auto-reload enabled.

TIM 14 handles the generation of 50% duty cycle PWM signals. The following formulas were used to calculate the counter period and pulse width:

$$\text{Counter Period} = \frac{\text{Clock Frequency}}{\text{Desiered PWM Frequency}} - 1 = \frac{16\text{MHz}}{1\text{kHz}} - 1 = 15,999 \text{ counts}$$

$$\text{Pulse Width} = 50\% \text{ of Counter Period} \approx 7,999$$

This sample calculation for a 1kHz signal can be applied to other target frequencies using the same formulas.

3. Menu Option Implementation and Results

The E-MultiTool provides five features: Angle Measure, Level Checker, Continuity Test, PWM Generator and free fall detection. A rotary encoder with an integrated switch manages menu navigation and selection. Pressing the switch once enters a mode, and pressing it again exits. This button press toggles the MCU between the MAIN_MENU and TASK_STATE. Once in the TASK_STATE, a switch statement executes the selected feature based on the encoder's current counter value. The implementation and results of these four functional modules are analyzed in the subsections below.

3.1 Angle Measure

The Angle Measure feature utilizes the on-board accelerometer to calculate the board's angle of inclination. The sensor provides a 10-bit output, mapping gravitational force across each axis from -255 to 255. By processing the raw values from the X and Z axes, the tilt angle is calculated using the formula:

$$\theta = \tan^{-1} \frac{X \text{ values}}{Z \text{ values}}$$

The inclination and its measured output are displayed in Figure 4.



Figure 4: Measured inclination

3.2 Level Checker

The Level Checker feature identifies surface flatness by monitoring the accelerometer's three axes. When the board lies perfectly flat, gravity is isolated to one axis, meaning the other two axes will measure between -10 and 10. This logic allows the device to detect levelness regardless of its initial orientation. The OLED represents these states using two distinct graphic indicators: a server carrying food appears when the surface is level, and a server spilling food appears when it is unlevel. Figures 5a and 5b illustrate these flat and non-flat visual states.

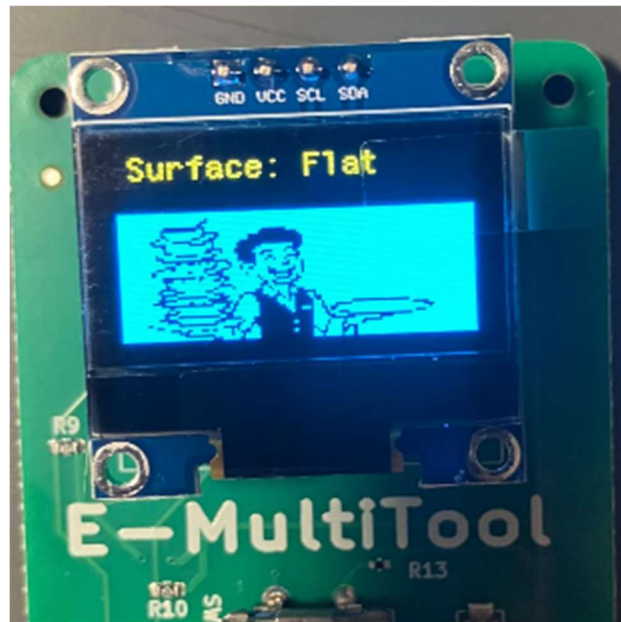


Figure 5a: Flat level indication

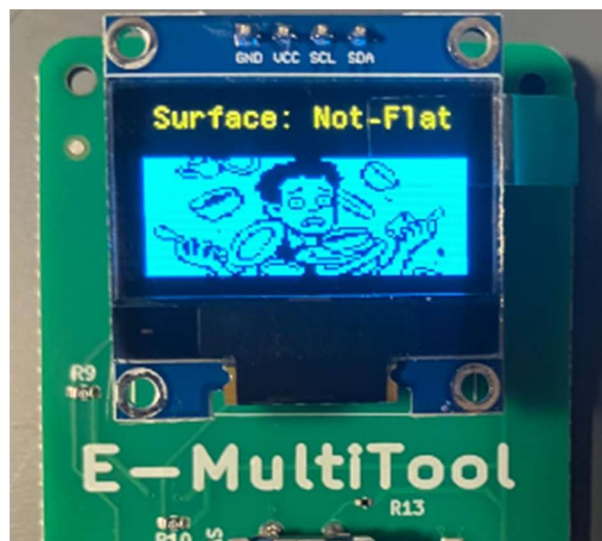


Figure 5b: Uneven surface indication

3.3 Continuity Test

The Continuity Test verifies circuit path conductivity using one of the MCU's GPIO pin that is connected to a 10k Ω pull-up resistor. To test a path, the user connects it between the test pin (TST) and GND. If the path is closed (continuous), the current takes the path of least resistance to ground, forcing the GPIO pin to a logic LOW state. The MCU detects this voltage change to confirm continuity. Figures 6a and 6b display the system behavior during discontinuity and continuity.

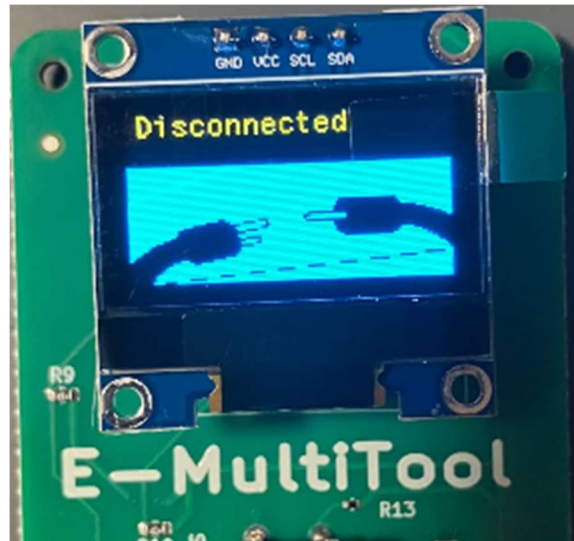


Figure 6a: Discontinuity indication



Figure 6b: Continuous indication

3.4 PWM Generator

The PWM Generator outputs four 50% duty-cycle signals, with the verified oscilloscope waveforms shown in Figures 7-10. The output frequency matches the target specifications, and the peak-to-peak voltage remains close to the expected 3.3 V. At higher operational frequencies, the pulse edges are rounded. This distortion is maybe caused by the hardware's inherent rise and fall time limits, which become more pronounced as the overall clock period decreases.

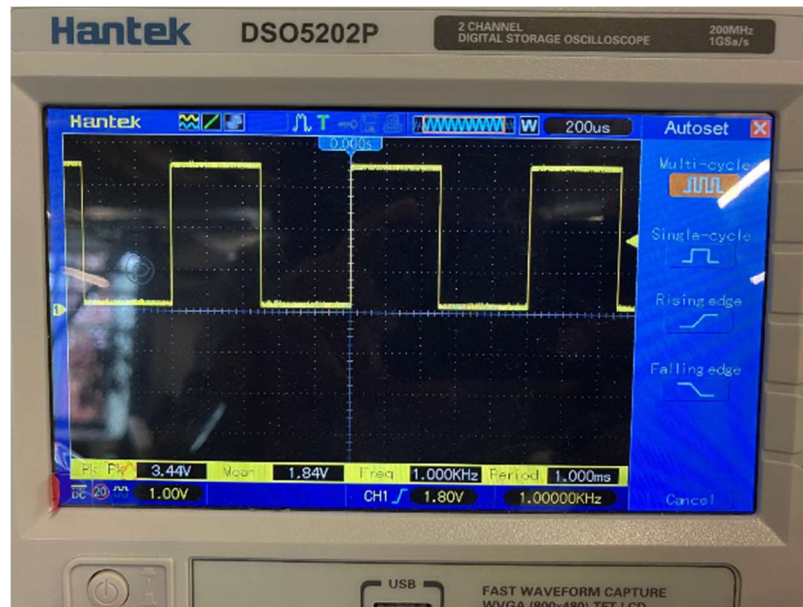


Figure 7: 1kHz PWM signal

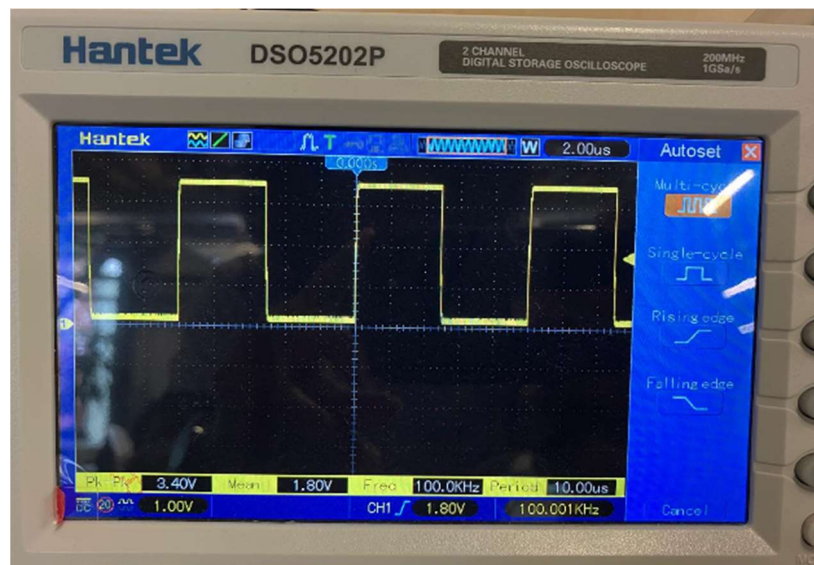


Figure 8: 100kHz PWM signal

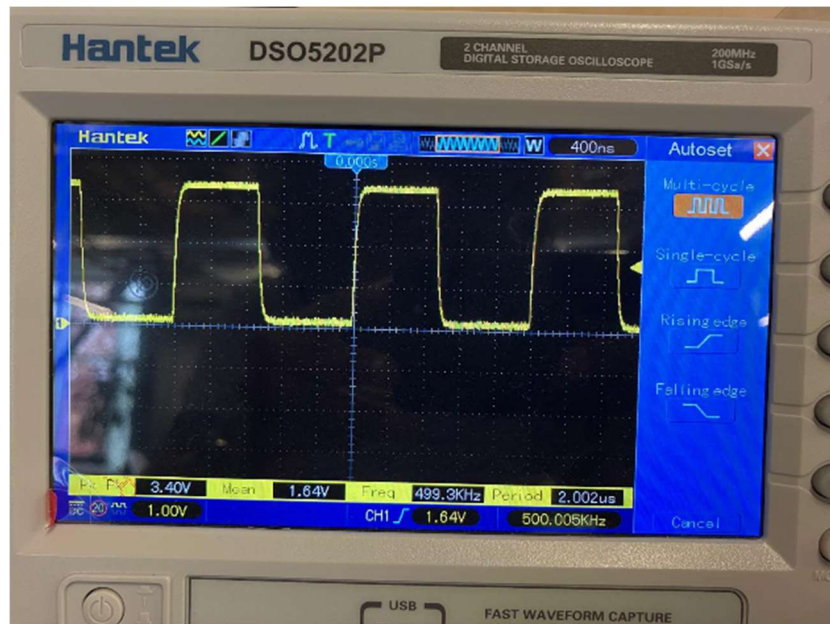


Figure 9: 500kHz PWM signal

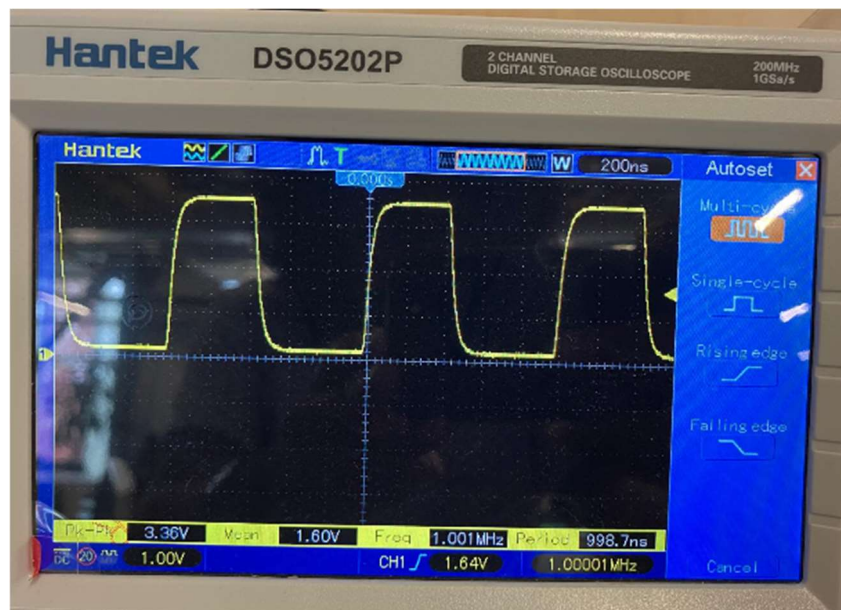


Figure 10: 1MHz PWM signal

3.5 Free Fall Detection

Free fall detection uses the onboard accelerometer to monitor weightlessness by configuring CTRL_REG3, INT1_CFG, and INT1_THS registers during initialization. A free fall state is confirmed when the acceleration on all three axes (X, Y, and Z) simultaneously drops below 320 mg. When this condition is met, the accelerometer interrupts the microcontroller and this interrupt is used to display a warning message on the OLED screen, as shown in Figure 11. The system remains in this state until the user presses the push-button to dismiss the message and return to the main menu.



Figure 11: Message displayed after free fall

4. Conclusion

The E-MultiTool successfully integrates embedded firmware with hardware design using an STM32F0 microcontroller. Testing confirmed that all four core features operate as expected, providing clear user feedback via the OLED display. Future hardware revisions can improve upon this design by adding onboard ESD protection and optimizing trace capacitance to minimize high-frequency signal distortion when generating a PWM signal.

5. Appendix

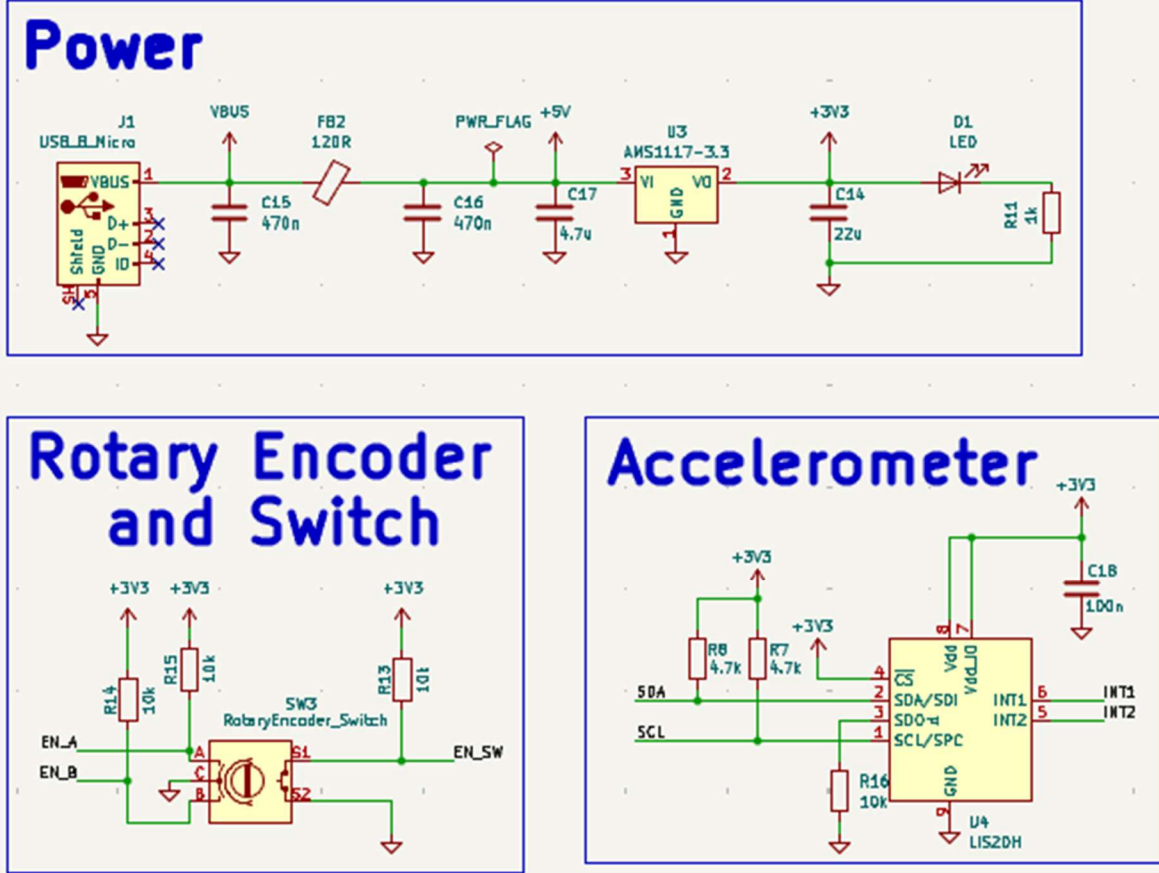


Figure A1: Power, Rotary encoder and Accelerometer schematic

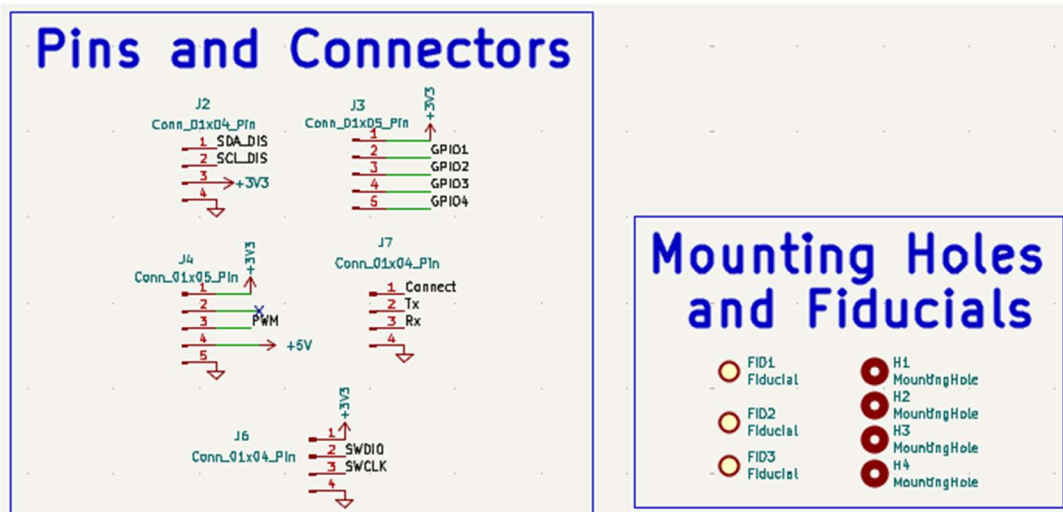


Figure A2: Pins, Connectors, Mounting Holes and Fiducials

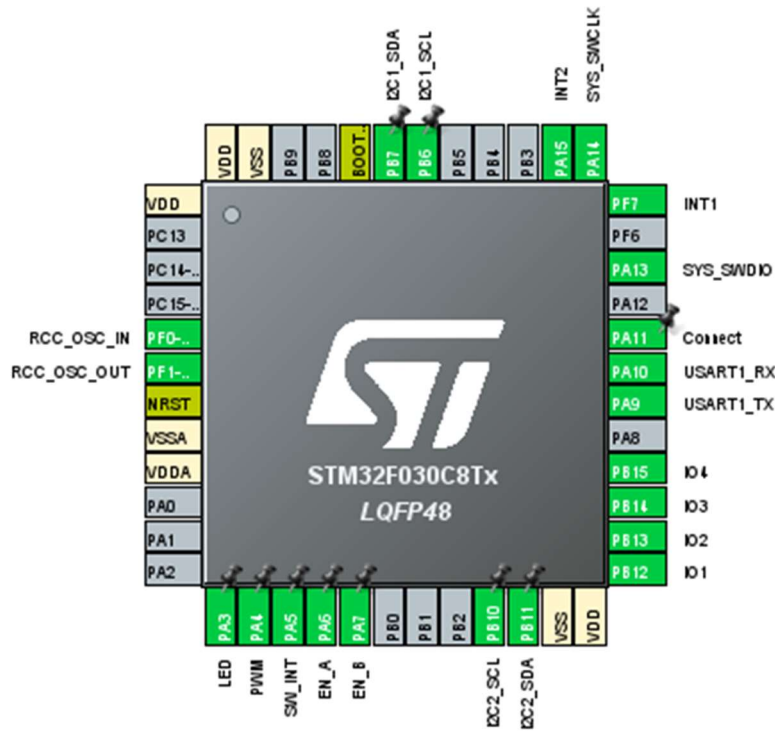


Figure A3: STM32F0 pin configuration from STM32 CubeIDE

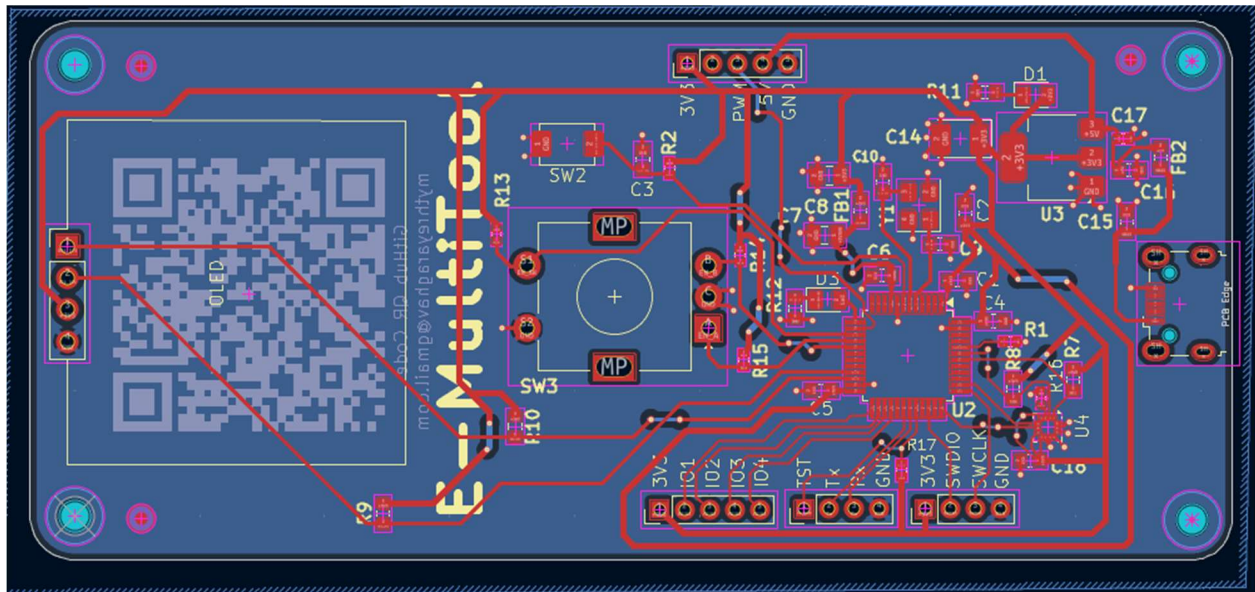


Figure A4: Routing of the E-MultiTool Board

*Note: The original driver code for the OLED display was written by Olivier Van den Eede. This code was then modified to fit my project requirements.

Original Code Link: <https://github.com/afiskon/stm32-ssd1306>